THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 9 & 10: **Cell: The Building Block of Life, Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The lifespan of a red blood cell is about: 1

(a) 12 days **(b)** 120 days **(c)** 1 year **(d)** 10 years

2. The ‘suicide bags’ of the cell, containing digestive enzymes, are: 1

(a) ribosomes **(b)** plastids **(c)** lysosomes **(d)** vacuoles

3. The colourless plastids that store starch, oil or proteins are: 1

(a) leucoplasts **(b)** chromoplasts **(c)** chloroplasts **(d)** mitochondria

4. Gametes (sex cells) are formed by: 1

(a) meiosis **(b)** binary fission only
(c) osmosis **(d)** mitosis

5. Genetically regulated cell death is called: 1

(a) plasmolysis **(b)** cancer
(c) mitosis **(d)** programmed cell death

6. The tissue that lines the lungs' air sacs and is very thin is: 1

(a) cuboidal epithelium **(b)** columnar epithelium
(c) ciliated muscle **(d)** squamous epithelium

7. Bones are connected to bones by: 1

(a) tendons **(b)** blood vessels **(c)** muscles **(d)** ligaments

8. Muscles attached to bones and used in running or writing are: 1

(a) nervous tissue **(b)** cardiac muscles
(c) involuntary smooth muscles **(d)** voluntary skeletal muscles

9. The joint at the knee and elbow is a: 1

(a) hinge joint **(b)** pivot joint
(c) ball and socket joint **(d)** fixed joint

10. The cartilage discs between the vertebrae: 1

(a) connect bones to muscles **(b)** produce blood
(c) store calcium **(d)** act as cushions and give flexibility

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is the nucleus called the control centre of the cell? 2

12. Why do mature human RBCs lack a nucleus? State one consequence. 2

13. Name the two conducting tissues in plants and their functions. 2

14. What are the functions of epithelial tissue? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why must a cell divide instead of growing indefinitely? 3

16. What are the components of blood? State its function. 3

17. Describe the stages of mitosis in a root tip briefly. 3

OR

Differentiate mitosis and meiosis (three points). 3

18. Compare bone and cartilage. 3

OR

Distinguish between voluntary and involuntary muscles with examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Compare prokaryotic and eukaryotic cells in four points.
(ii) Give one example of each.
(iii) Where is the DNA of a bacterium found? 5

OR

(i) What is the cell cycle?
(ii) Compare mitosis and meiosis.
(iii) State the importance of cell division. 5

20. **Case study:** Different muscles in the body: biceps, heart, intestine. 5
(i) Classify each.
(ii) Which are voluntary?
(iii) Which is striated and branched?
(iv) State one function of each.

OR

(i) Describe skeletal muscle, smooth muscle and cardiac muscle.
(ii) Differentiate voluntary from involuntary movements. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND